REVERSE ENGINEERING – ACTIVITY REPORT

Product: EVOFOX Wired Optical Gaming Mouse

Date	09 October 2026
Institution	B.Tech – KIET University
Group Members	Shourya Pratap Singh, Arnav Mishra, Tarun Singh Rana, Tanishq Sharma



Fig. 0 EVOFOX wired gaming mouse (product studied)

PART I – STUDYING THE PRODUCT

1. External Features

Physical construct: White moulded plastic shell (top cover and base) with two main buttons, a scroll wheel, a button below the wheel (typically DPI select), side buttons, a braided USB cable with a ferrite noise-suppression core and a USB-A plug, glide feet and an optical window on the underside.

Subsystems used: (a) optical tracking sensor with LED and lens, (b) microcontroller / USB interface, (c) click buttons (microswitches), (d) scroll wheel with encoder, (e) power and cable.

Description: An LED lights the surface and a CMOS sensor captures images to detect movement; the controller combines this with button and wheel inputs and sends it to the computer over USB.

Modes of operation: active tracking, idle/sleep when stationary (reduces power), and DPI selection on models with a DPI button.

2. Purpose of the Product

To act as a pointing input device that converts hand movement, clicks and scrolling into digital signals so the user can control the cursor and interact with a computer.

3. User Needs Addressed

Precise and smooth cursor control; quick selection through clicks; easy scrolling; comfortable ergonomic shape; plug-and-play use with no driver; works on most surfaces; low cost and durability.

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

Stage	Technology	Improvement / drawback
1.1 Early (1960s)	Wooden mouse with two perpendicular wheels (Engelbart)	First pointing device; bulky, hard to use
1.2 First generation	Mechanical ball mouse with rollers and slotted encoder discs	Compact, practical; ball and rollers collect dirt, need cleaning

Stage	Technology	Improvement / drawback
1.3 Second generation	Early optical mouse needing a special gridded pad	No moving parts; required special mousepad
1.4 Third generation	LED optical mouse with CMOS image sensor (late 1990s) – this type	Works on most surfaces, no cleaning; struggles on glass or glossy surfaces
1.5 Latest	Laser and high-DPI optical sensors, wireless 2.4 GHz / Bluetooth, rechargeable, programmable gaming mice	Very accurate on any surface, cable-free; costlier, needs battery

2. Current Status – Core Smart Technologies

CMOS optical image sensor with on-chip DSP (thousands of frames per second); USB HID class for driver-free operation; optical or mechanical scroll encoder; microswitches rated for millions of clicks; power-saving sleep mode; in modern models, wireless links and adjustable DPI.

3. Engineering Importance of This Evolution

The mouse evolved from a mechanical to a fully solid-state device, showing how image processing, embedded microcontrollers and standard interfaces (USB HID) can replace moving parts, improving accuracy and reliability while reducing cost. It is a clear example of sensor-based motion detection used widely in robotics and navigation.

PART III – INTERNAL COMPONENTS AND WORKING FLOW

1. Introduction of the Product

A wired optical mouse built around a CMOS optical sensor and USB controller on a single small PCB. Components are identified from their markings; values marked “typical” should be confirmed on the actual product.

2. Photograph of Internal Components

[Insert photograph of the opened mouse showing PCB, sensor, LED, lens, switches and scroll wheel]

3. Internal Block Diagram (Functional View)

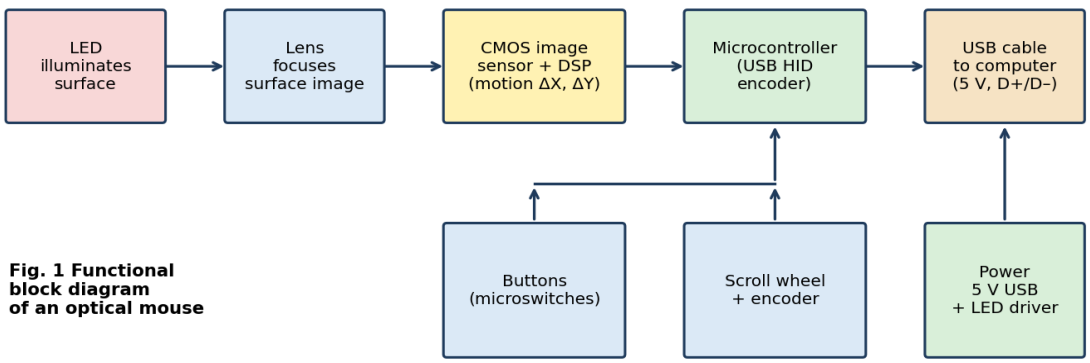


Fig. 1 Functional block diagram of an optical mouse

4. Internal Subsystems with Detailed Explanation

(4.1 function, 4.2 working principle, 4.3 characteristics, 4.4 role in product)

Subsystem	4.1 Function	4.2 Working principle	4.3 Characteristics	4.4 Role
LED + lens	Illuminate and focus the surface	LED lights the surface at a low angle so texture casts shadows; lens focuses it onto the sensor	Red LED ~1.8–2 V, 10–20 mA	Provides image to track
Optical sensor (CMOS + DSP)	Detect motion	Captures successive images (small pixel array); DSP compares them to find ΔX and ΔY	~800–1600 DPI; thousands of frames/s	Core of tracking
Microcontroller / USB interface	Process and send data	Combines motion, button and wheel data into a USB HID report	5 V, low-speed USB; 125 Hz poll rate typical	Communicatio n with computer
Microswitches	Detect clicks	Metal leaf snaps and closes contact when button is pressed; debounced in firmware	~5–10 million clicks life	Click input
Scroll wheel + encoder	Detect scrolling	Rotation interrupts or switches two signals (quadrature) so direction is detected	Two-phase output, notched wheel	Scroll input
USB cable & power	Supply power and carry data	4 wires: +5 V, D+, D–, GND	5 V, about 100 mA	Power and data link

5. Complete Working Flow (Step-by-Step)

1. The mouse is plugged in; the computer supplies 5 V and detects it as a HID device.
2. The LED illuminates the surface and the lens focuses its texture onto the CMOS sensor.
3. The sensor captures images thousands of times per second and the DSP compares them to calculate ΔX and ΔY.
4. Button presses close the microswitches and the scroll encoder produces direction pulses; these are read by the controller.
5. The controller packs movement, button state and scroll data into a USB report.
6. The computer driver reads the report and moves the cursor, registers clicks and scrolls the page.

6–8. Bill of Material, Ratings and Function Table

Component Name	Qty	Function	Specifications (typical – verify)
Optical sensor IC	1	Detects movement	CMOS, 800–1600 DPI
LED (red)	1	Illuminates surface	~1.8 V, 20 mA
Lens / light guide	1	Focuses image	Plastic moulded
Microcontroller / USB chip	1	Processes and sends data	5 V, USB HID
Microswitches	2–3	Detect clicks	~5–10 M clicks
Scroll encoder + wheel	1	Detects scrolling	Quadrature, 2-phase
Resistors, capacitors	Sever al	Current limiting, decoupling	0.1 µF, 10 µF, 100–330 Ω
USB cable + connector	1	Power and data	4-core, 5 V

Component Name	Qty	Function	Specifications (typical – verify)
PCB and plastic housing	1	Mounting and protection	FR4 / ABS

PART IV – SIMULATION AND RESULTS

Items 2–4 to be repeated for every participant.

1. Circuit Diagram

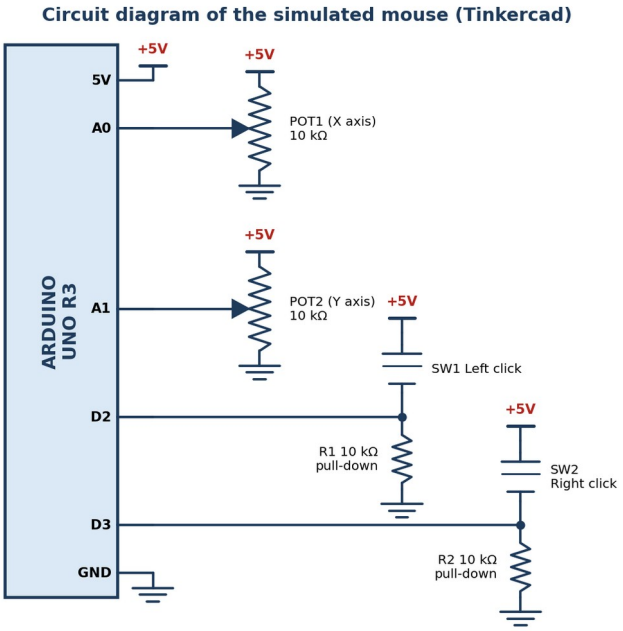


Fig. 2 Circuit diagram of the simulation (Arduino Uno, 2 potentiometers for X/Y, 2 pushbuttons with 10 kΩ pull-down resistors)

2. Screenshot of Simulation Circuit

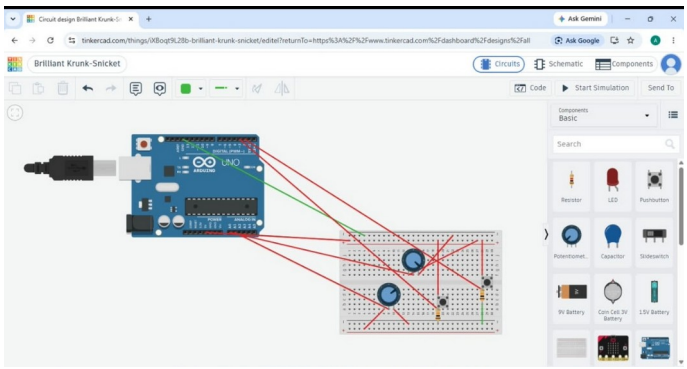


Fig. 3 Tinkercad simulation circuit (Arduino Uno, two potentiometers, two pushbuttons)

[Other participants: insert your simulation screenshots here]

3. Video Link

[Participant name: _____ GitHub video link: _____]

4. Simulation Code

Arduino Uno code (Tinkercad) for a button and rotary encoder, modelling the click and scroll inputs of the mouse. Connect encoder A/B to pins 2 and 3, button to pin 4 (to GND) and an LED to pin 13. Untested starting point.

```
const int PIN_A = 2, PIN_B = 3, PIN_BTN = 4, PIN_LED = 13;
int lastA;
long scrollCount = 0;
bool lastBtn = HIGH;

void setup() {
  pinMode(PIN_A, INPUT_PULLUP);
  pinMode(PIN_B, INPUT_PULLUP);
  pinMode(PIN_BTN, INPUT_PULLUP);
  pinMode(PIN_LED, OUTPUT);
  lastA = digitalRead(PIN_A);
  Serial.begin(9600);
}

void loop() {
  int a = digitalRead(PIN_A);
  if (a != lastA) {            // encoder moved
    if (digitalRead(PIN_B) != a) scrollCount++;
    else scrollCount--;
    Serial.print("Scroll: ");
    Serial.println(scrollCount > 0 ? "UP" : "DOWN");
    lastA = a;
  }
  bool btn = digitalRead(PIN_BTN);
  if (btn != lastBtn) {        // click detected
    delay(20);                 // debounce
    btn = digitalRead(PIN_BTN);
    if (btn == LOW) {
      Serial.println("Left click");
      digitalWrite(PIN_LED, HIGH);
    } else {
      digitalWrite(PIN_LED, LOW);
    }
    lastBtn = btn;
  }
}
```

Expected result: scroll direction and "Left click" appear in the Serial Monitor.

Submit: this PDF on the LMS; all group videos on GitHub.